

Create Professional PCB Designs with ZenitPCB Suite

A free, easy-to-use computer-aided design software package for Windows platform, that lets you design printed circuit boards of professional quality very quickly

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Several electronic design automation (EDA) software are available in the market, which provide professional-grade printed circuit board (PCB) designs. While the premium and full-featured versions of design software can be pretty expensive, there are several alternatives in the form of free integrated design environments (IDEs) available these days.

ZenitPCB is one such excellent EDA toolset that allows PCB designing for electronics projects. This freeware is flexible, has a simple user interface (UI) and lets you realise the design in a short period of time.

This design suite comprises four tools: ZenitPCB Layout, ZenitCapture Schematic, ZenitPCB Parts and ZenitPCB GerberView.

Why ZenitPCB

The ZenitPCB software package is absolutely free of cost and still provides designs with professional levels of accuracy. It comes with some advanced features offered by paid software—extensive and well-managed library, multiple layers and double-sided designs support, high flexibility, superior schematic design, compatibility with standard design file formats and a tool for viewing Gerber file.

The software reduces the time to market; for example, the user can import a mechanical drawing with a PCB and build his/her board outline, thereby avoiding mistakes and saving time.

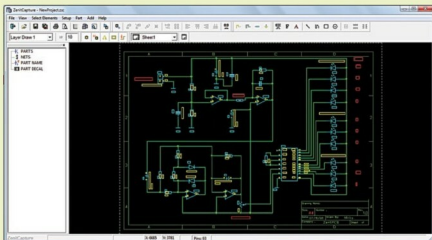


Fig. 1: Screenshot of ZenitCapture project

Who can use it

ZenitPCB Suite and the accompanying libraries are licensed as freeware and can be used for personal, educational and semi-professional purposes. It is aimed at designers, engineers, hobbyists, students and academicians who want to create their own PCBs with a professional touch, without having to pay for expensive licences. The circuit board for design has 800 pins and serves the level of complexity that would be required by designs for the targeted audience. This 800 pins limit, according to the developer, is the borderline between the hobby projects and professional jobs.

Designing schematics with ZenitCapture

Schematic designs can be created quickly and easily using the feature-rich toolset of ZenitCapture.

Schematic is a diagram that acts like a map for designing, building and troubleshooting electronic or electrical circuits. It gives us the plan of components used for a design (represented as electronic symbols) and how they are connected.

A schematic in ZenitCapture can have several layers and the layout is created in a netlist file format. The part libraries have a collection of more than 800 commonly-used components (both surface-mount devices and through holes) suitable for immediate placement on the project. The device required for design can be either selected from these libraries or edited/created using ZenitParts.

Connections can be easily added with orthogonal locking and snap-to-pin features. You can also place global symbols (ground pin - GND and power supply pin - Vcc), IEEE symbols